



Python Full Stack Developer Program

Master backend programming with Python and Flask, and frontend development with HTML, CSS & JavaScript — backed by hands-on projects.

LEARNING PATH · PROGRAM SYLLABUS · REAL-TIME PROJECTS & CASE STUDIES

100 Days

Classroom + Live Training

31

Curriculum Modules

9

Hands-On Projects

30k+

Students Trained

01 Learning Path

1

Python Programming Fundamentals

Phase 1

Setting up the Python environment, syntax & data types, operators, strings, tuples, lists, sets, dictionaries, conditional statements, loops, and list/dictionary comprehension.

2

Functions, OOP & Core Libraries

Phase 2

Functions, lambda & generators, modules and packages, exception handling, Object-Oriented Programming (classes, inheritance, polymorphism), date/time, regex, and file handling.

3

Database Programming — MySQL

Phase 3

Database fundamentals, SQL sub-languages, operators, joins, working with MySQL Workbench, and aggregate functions for building data-driven apps.

4

Backend Web Development — Flask

Phase 4

Flask architecture and installation, virtual environments, building dynamic routes, request/response handling, and the Jinja2 templating engine.

5

Frontend Development & Deployment

Phase 5

HTML page structure & forms, CSS3 core & layouts, JavaScript fundamentals, DOM manipulation, ES6 concepts, and cloud deployment.



02 Program Syllabus

TERM 1 · PYTHON CORE PROGRAMMING

CORE

Syntax, Data Types & Operators

- Python I/O operations, variables & comments
- Basic data types and typecasting
- Arithmetic, assignment, comparison & logical operators
- Identity, membership & bitwise operators

Strings, Collections & Comprehension

- String creation, formatting, indexing & slicing
- Tuples, lists, sets — properties and built-in methods
- Dictionaries — keys, values & methods
- List and dictionary comprehension

Control Flow & Functions

- If, if-else & if-elif-else statements
- While & for loops, break, continue, enumerate, zip
- Functions — positional, keyword & variable-length arguments
- Lambda functions, generators, and recursion

OOP, Modules & Exception Handling

- Classes, objects, constructors & access modifiers
- Inheritance, polymorphism & operator overloading
- Creating modules & packages; built-in modules
- Try/except/finally, user-defined exceptions

Date, Regex & File Handling

- Date, time & datetime modules, strftime/strptime
- Regular expressions — search, compile, split, sub
- Opening, reading, writing & closing files
- HackerRank practice for bronze/silver-level badges

TOOLS & TECHNOLOGIES

Python

PyCharm/VS Code

Git

HackerRank

TERM 2 · DATABASE & BACKEND DEVELOPMENT

BACKEND

MySQL Database

- Introduction to databases & SQL sub-languages
- MySQL operators and comparison operators
- DDL: alter and rename, constraints, string functions
- Aggregate functions, SQL files & MySQL joins

Python Web Development — Flask

- Flask architecture, installation & virtual environments
- Building dynamic routes and redirection
- URL building, URL converters, request & response handling
- Server-side sessions and file handling with Flask

TOOLS & TECHNOLOGIES

MySQL

MySQL Workbench

Flask

Postman



TERM 3 · FRONT-END WEB DEVELOPMENT

FRONTEND

HTML & CSS3

- Basic page structure, formatting page content
- Creating lists, links & structuring content
- CSS3 core, typography & layouts
- Dynamic web pages with Flask-Jinja2

JavaScript for Full-Stack Apps

- JavaScript basics, control flow & arrays
- Functions, loops & essential built-in methods
- JavaScript and the DOM
- ES6 concepts and cloud deployment

TOOLS & TECHNOLOGIES

HTML5

CSS3

JavaScript

Bootstrap



03 Real-Time Projects & Case Studies

Instagram Database Clone Project

Model Instagram's core data relationships and build the underlying database schema and queries.

Hands-on Build

Custom DevBlog Application Project

A Flask-powered blogging platform with article creation, editing and category/tag filtering.

Hands-on Build

Internal Template (HTML/CSS) Project

Build a reusable internal site template practicing semantic HTML and CSS layout techniques.

Hands-on Build

Instagram Login Page Project

A pixel-accurate clone of the Instagram login interface, focused on form validation and styling.

Hands-on Build

Fully Functional E-Commerce App Capstone

Browse products, manage cart/wishlist, and complete checkout flows backed by a Flask + MySQL backend.

Capstone Project

Calculator Using JavaScript Project

A fully interactive calculator built with vanilla JavaScript DOM manipulation and event handling.

Hands-on Build

Random Password Generator Project

Generate secure, customisable passwords with adjustable length and character-set rules.

Hands-on Build

Discussion Board Application Project

A threaded discussion forum with posts, replies and category-based navigation.

Hands-on Build

Personal Records Manager Project

A CRUD application to create, update, and organise personal records backed by a MySQL database.

Hands-on Build